

Cloud Fundamentals

1 Cloud Fundamentals

1. Cloud computing delivers computing services over the internet on demand.
2. Cloud eliminates the need to manage physical servers onsite.
3. Cloud shifts IT from capital expenditure (CapEx) to operational expenditure (OpEx).
4. Pay-as-you-go pricing reduces upfront infrastructure cost.
5. Elasticity allows automatic scaling based on workload.
6. On-demand self-service enables users to provision resources without manual intervention.
7. Cloud provides global accessibility and faster deployment.
8. Cloud improves agility and time-to-market.
9. Cloud supports seasonal traffic spikes without hardware purchase.
10. Public cloud reduces infrastructure maintenance burden.
11. Cloud adoption improves scalability and operational efficiency.

2 Cloud Deployment Models

1. Public cloud provides shared infrastructure managed by a provider.
2. Private cloud is dedicated to a single organization.
3. Hybrid cloud combines on-premises and public cloud environments.
4. Community cloud serves organizations with shared requirements.
5. Hybrid cloud is ideal for keeping sensitive data onsite while using public services.

3 Cloud Service Models

1. IaaS provides virtualized compute, storage, and networking.
2. PaaS allows application deployment without managing OS or runtime.
3. SaaS delivers ready-to-use software over the internet.
4. PaaS suits developers building custom applications.
5. SaaS suits users needing ready-to-use applications.

6. Cloud management platforms centralize private cloud control.

Cloud Security & Governance

1. Cloud security follows the Shared Responsibility Model.
2. Providers secure infrastructure; customers secure data and access.
3. IAM controls user permissions and roles.
4. Role-Based Access Control (RBAC) enforces least privilege.
5. MFA strengthens authentication security.
6. CloudTrail enables activity logging and auditing.
7. AWS Config tracks resource configuration compliance.
8. Encryption at rest protects stored data.
9. Cloud security includes compliance, governance, and cost management.
10. Multi-tenancy requires strict identity and access control.

5 **AWS Core Concepts**

1. AWS account creation is required before accessing services.
2. AWS Management Console is the primary web interface.
3. EC2 provides virtual servers.
4. Instance types define CPU and memory capacity.
5. Security Groups control inbound and outbound traffic.
6. SSH (Linux) and RDP (Windows) enable remote access.
7. VPC provides isolated networking.
8. Elastic IPs provide static public IP addresses.
9. Regions provide geographic separation.
10. Availability Zones provide high availability within a region.
11. Multi-AZ deployments improve resilience.
12. Auto Scaling adjusts capacity automatically.
13. Elastic Load Balancer distributes traffic.

14. CloudFront delivers content globally with low latency.
15. Route 53 provides DNS services.
16. S3 provides scalable object storage.
17. S3 Glacier supports archival storage.
18. Lifecycle policies move data to cheaper storage tiers.
19. S3 versioning protects against accidental deletion.
20. Default bucket encryption secures stored data.
21. Temporary credentials allow limited-time access.

6 AWS Architecture & High Availability

1. High availability requires deployment across multiple Availability Zones.
2. Global deployment requires multiple Regions.
3. VPC enables subnet isolation and network segmentation.
4. Security Groups must allow required ports (e.g., port 22 for SSH).
5. Compute-optimized instances support high-CPU workloads.
6. Load balancing increases reliability and scalability.
7. Auto Scaling improves cost efficiency.

7 Azure Core Concepts

1. Azure Subscription manages billing and service access.
2. Azure Resource Groups organize related resources.
3. Azure Portal is used for management and monitoring.
4. Azure Virtual Machines provide compute resources.
5. Azure Marketplace simplifies VM deployment.
6. Virtual Network (VNet) enables private networking.
7. VNet Peering connects VNets securely.
8. Availability Sets improve VM redundancy.
9. Azure Blob Storage provides scalable object storage.

10. Azure SQL Database supports structured data workloads.
11. Azure Backup enables automated VM backup.
12. Azure Cost Management tracks spending across subscriptions.

8 Storage & Backup Concepts

1. Object storage supports scalable, redundant data storage.
2. Archival storage reduces cost for infrequently accessed data.
3. Backup policies define retention periods.
4. Lifecycle management optimizes storage cost.
5. Versioning protects against accidental deletion.
6. Encryption ensures data protection at rest.

9 Cloud Operational Challenges

1. Cloud migration introduces security and compliance risks.
2. Multi-cloud environments increase operational complexity.
3. Cost management is a key cloud governance challenge.
4. Performance monitoring is essential in cloud environments.
5. Proper resource configuration ensures resilience and security.